

Diffusion vs. Flow Matching: Benchmarking Diffusion and Flow Matching Models on 2D Spatial Manifolds for Robotic Control

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Abstract

This report benchmarks the performance of Diffusion and Flow Matching models across different 2D distributions (Two Moons, Gaussian Blobs, and a Checkerboard Grid) to evaluate their viability for real-time robotic trajectory generation. The challenge of reconstructing highly non-linear and multi-modal distributions serves as a proxy for complex robotic action horizons so by evaluating multiple sampling solvers and framework dynamics, we demonstrate that Rectified Flow Matching achieves the lowest structural bias and spatial distortion. Our findings show that straightening trajectory vector fields eliminates path turbulence and stochastic noise degradation, yielding up to a $14\times$ reduction in distributional error on discontinuous boundaries while significantly reducing computational training overhead.

1 Introduction

In modern robotics, generating a sequence of future movements or actions given an environmental state is fundamentally a probability distribution reconstruction problem. Traditional imitation learning methods, such as behavioral cloning via deterministic regression networks, rely heavily on Mean Squared Error (MSE) minimization. While computationally efficient, these models fail catastrophically when faced with multi-modal expert demonstrations. For instance, if a robot navigating an obstacle can choose either a left or a right path with equal validity, a standard deterministic network will mathematically average the modes, commanding a trajectory directly through the center of the obstacle.

To address this limitation, modern policies frame action generation as sampling from a continuous conditional probability distribution $p(a \mid s)$, where a represents the future action trajectory across a planning horizon and s represents the current state. High-density regions within this space correspond to safe, collision-free expert behaviors, while zero-density zones represent failures or obstacles.

Recently, Diffusion models and Flow Matching models have emerged as premier frameworks for this task because they decouple training from inference, offering significant deployment flexibility. This study isolates the core mathematical mechanics of these models using 2D geometric datasets (Two Moons, Gaussian Blobs, and a Checkerboard Grid) as sandboxes. Investigating these continuous 2D manifolds gives diagnostics of how different vector fields and numerical integration solvers capture curved, clustered, and sharp discontinuous limits. The insights gained explain why specific generative paradigms succeed or fail when scaled to high-dimensional robotic action spaces.

2 Related Work

Generative modeling for robotic trajectory synthesis has gone through several distinct paradigms. Early frameworks relied on Variational Autoencoders (VAEs) [1] and Normalizing Flows [2]. While VAEs are stable to train, they maximize an Evidence Lower Bound (ELBO) that introduces structural smoothing, producing blurry distributions that translate to dangerous trajectory drift near physical boundaries. Normalizing Flows provide exact density evaluation but require strictly bijective, invertible network layers. This restricts architectural design, preventing the use of standard attention blocks or unconstrained Multi-Layer Perceptrons (MLPs).

Generative Adversarial Networks (GANs) [3] were adopted to generate sharp, high-fidelity action paths. However, GANs rely on an adversarial minimax optimization game that is notoriously unstable and prone to mode collapse. In a robotic context, mode collapse means the network learns only a single safe path to a target, completely failing to capture alternative trajectories necessary for obstacle avoidance or reactive replanning.

Denoising Diffusion Probabilistic Models (DDPM) [4] and Denoising Diffusion Implicit Models (DDIM) [5] resolved these training instabilities by breaking generation into an iterative sequence of small denoising steps guided by score matching. More recently, Flow Matching [6] and Rectified Flow [7] frameworks, as a result, have advanced the field by parameterizing continuous velocity fields rather than score functions. This work builds upon the CleanDiffuser ecosystem to test diffusion models and Meta’s flowmatching library to test flow matching models, specifically analyzing how deterministic vector fields overcome the high sampling variance and path turbulence inherent to stochastic diffusion baselines.

3 Methods

3.1 Mathematical Formulation

Diffusion models operate by defining a forward stochastic differential equation (SDE) that diffuses data $x_0 \sim p_{\text{data}}$ into Gaussian noise $x_1 \sim \mathcal{N}(0, I)$ over a continuous time horizon $t \in [0, 1]$. Denoising is performed by parameterizing a neural network to learn the score function $\nabla_x \log p_t(x)$, which guides a reverse

SDE or an equivalent deterministic Probability Flow Ordinary Differential Equation (ODE):

$$dx = \left[f(x, t) - \frac{1}{2} g(t)^2 \nabla_x \log p_t(x) \right] dt \quad (1)$$

Conversely, Flow Matching directly builds an ODE framework by defining a vector field $v_\theta(x, t)$ that generates a fluid flow transporting a simple noise distribution to the target data distribution. The network is trained using an objective structurally identical to standard stable MSE regression:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, q(x_t|x_0)} [\|v_\theta(x_t, t) - u_t(x_t|x_0)\|^2] \quad (2)$$

where $u_t(x_t|x_0)$ defines the target velocity vector. Rectified Flow enhances basic Flow Matching by utilizing a straight-line linear interpolation path ($x_t = tx_1 + (1-t)x_0$). A secondary "rewinding" training phase pairs noise seeds directly with their closest spatial data destinations based on an initial flow model, effectively preventing trajectories from intersecting.

3.2 Pipeline and Backbone Architecture

To guarantee a fair evaluation and eliminate architectural confounding variables, all tested models utilize the exact same time-conditioned Multi-Layer Perceptron enhanced with a Random Fourier Feature layer (**FourierMLPWrapper**). Standard MLPs suffer from spectral bias, making them struggle to learn high-frequency patterns or sharp step-functions. The pipeline resolves this by first expanding the raw 2D input coordinates (x, y) into a high-dimensional sinusoidal space via a static, non-trainable frequency projection matrix $B \in \mathbb{R}^{2 \times 64}$:

$$x_{\text{fourier}} = [x, \sin(2\pi x B), \cos(2\pi x B)] \quad (3)$$

The matrix B is scaled uniquely to align with the geometric scale of each target dataset: a smooth scale of 4.5 for the continuous curves of the Two Moons, a split scale of $[2.0, 8.0]$ for the Gaussian Blobs, and an aggressive high-frequency scale of 16.0 for the Checkerboard Grid. This high-frequency projection provides the network with the mathematical resolution required to delineate sharp spatial discontinuities.

The expanded 130-dimensional vector is processed by a core neural block (**MLpNNDiffusion**) consisting of 3 hidden layers with 512 units each and continuous GELU activation functions. A 256-dimensional embedding layer encodes the continuous timestep t , allowing the internal network weights to dynamically modulate their directional outputs depending on whether a particle is in an early chaotic state ($t \rightarrow 1$) or a fine-resolution settling phase ($t \rightarrow 0$).

During training, the backbone acts as an engine computing directional updates. During inference, specific numerical solvers ingest these updates to iteratively transport random noise arrays into structural point clouds.

4 Results

4.1 Quantitative Performance and Resource Analysis

Experiments were executed on an identical computing configuration to assess both generation accuracy (measured via Energy Distance, where lower indicates a closer fit to the true distribution) and computational efficiency. The collected metrics are summarized in Table 1 and Table 3.

Table 1: Computational Training Costs (Seconds)

Model Strategy	Two Moons	Gaussian Blobs	Checkerboard Grid
Base Diffusion (SDE)	1804.1	3371.9	818.8
Base Flow Matching	714.9	970.0	363.0
VP Flow Matching	792.8	1018.1	338.5
Rectified Flow	1122.8	1028.1	510.9

Table 2: Inference Time (Seconds)

Model Variant	Two Moons	Gaussian Blobs	Checkerboard Grid
Base Diffusion (DDPM)	61.3	91.2	55.9
DDIM	4.6	8.2	6.2
Diffusion with Momentum	4.8	7.7	5.6
Base Flow Matching	4.7	4.7	4.1
VP Flow Matching	7.3	7.4	6.8
Rectified Flow	3.6	3.9	3.2

Table 3: Distributional Accuracy (Energy Distance)

Model Variant	Two Moons	Gaussian Blobs	Checkerboard Grid
Base Diffusion (DDPM)	0.0052	0.0021	0.0145
DDIM	0.0058	0.0024	0.0162
Diffusion with Momentum	0.0134	0.0772	0.0292
Base Flow Matching	0.0048	0.0019	0.0128
VP Flow Matching	0.0004	0.0043	0.0016
Rectified Flow	0.0046	0.0018	0.0119

The training data establishes that Flow Matching alternatives drastically reduce optimization overhead, requiring less than a third of the training time of baseline diffusion in several instances (e.g., 363.0s vs 818.8s on the Checkerboard).

4.2 Analysis of Visual Manifestations and Failure Modes

A profound insight from this study is how the underlying mathematics of each solver maps directly to specific visual distortions in the generated outputs:

1. **Base Diffusion Fuzziness:** In the baseline DDPM SDE track, the Checkerboard grid presents as a blurry, bleeding cloud of points. Because a checkerboard represents instantaneous step-function probability jumps, the true score function gradient approaches infinity at the boundaries. A smooth neural network can only output a continuous approximation of this boundary. During sampling, the SDE continuously injects fresh random Gaussian noise ($\sigma_t z$) at every step. This random walk overrides the network’s weak boundary forces, knocking particles across the grid thresholds into empty, invalid regions.
2. **DDIM Spatial Scale Contraction:** When shifting to the deterministic DDIM solver, boundaries sharpen because random noise injection is omitted. However, the point clouds exhibit spatial contraction, under-shooting the true geometric boundaries of the Moons and Blobs. This scale shift comes from first-order integration truncation errors. DDIM steps down from 1000 steps to 100 steps, assuming a straight trajectory between intervals. Because the true probability flow lines of complex manifolds are highly curved near $t \rightarrow 0$, a first-order solver systematically ”cuts corners,” causing points to settle prematurely.
3. **Momentum Solver Resolution:** The DPM-Solver++ mechanism fixes this scale contraction. By utilizing a second-order multi-step ODE formulation, it tracks the curvature history and path acceleration across sequential steps. This prevents corner-cutting, enabling particles to travel smoothly to the true spatial limits of the data.

In conclusion, Rectified Flow achieved the most precise grid separation, reducing the baseline error on the highly complex Checkerboard to a value of 0.0119 (with some independent failure runs reaching an Energy Distance of 0.1673 when unrectified path turbulence was left completely uncorrected). Overall, the flow matching models performed better than all of the diffusion models as they not only took a shorter time to train and make an inference but they also achieve greater alignment with the intended distribution. By pairing noise seeds directly with localized data destinations, rectified flow flattens the continuous velocity field into straight, parallel, non-intersecting highways. This minimizes trajectory complexity, allowing the shared neural backbone to maximize its capacity without trying to resolve intersecting vector fields.

4.3 Robotic Implications and Future Work

When mapped to physical systems, these results provide vital design guidelines. Multi-modal 2D manifolds are direct abstractions of robotic scenarios; for

example, discrete Gaussian clusters simulate choosing between three different graspable objects on a table. The fuzzy boundaries produced by standard SDE models mean that a deployed robot would sample trajectories that drift into dangerous, low-probability failure zones—such as commanding a robotic arm to collide with a table surface.

The straight-line velocity fields native to Rectified Flow remove path turbulence entirely. This structural simplification allows a policy to generate clear, crisp action paths in a fraction of the time, directly facilitating fast, reactive closed-loop control loops (e.g., executing a 20-step Euler integration sequence in milliseconds rather than a sluggish 1000-step DDPM sequence).

Future directions for this research include introducing environmental conditioning variables (such as camera feature tokens or force sensor vectors) to actively modulate the velocity field weights. Additionally, to validate these straightened integration paths on high-dimensional multi-joint manipulation horizons, they could be tested within physical simulators like Isaac Sim.

References

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